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Powder coating nozzle

Abstract

A paint nozzle may include a nozzle body defining a nozzle body axis. The nozzle body includes one or more sidewalls, an engagement upper surface and a dispensing lower surface. An internal nozzle chamber is formed between the upper and lower surfaces. A center dispensing channel connects the internal nozzle chamber to the dispensing lower surface and is parallel and coincident with the nozzle body axis. Two or more vortex dispensing channels are spaced axially outward from the center dispensing channel and are orientated at a dispensing angle relative to the nozzle body axis in order to generate a vortex flow around the center dispensing channel.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to a nozzle for a paint application system. In particular, the present invention relates to a paint nozzle for delivering powdered paint in a controlled vortex from a compact nozzle utilizing a triboelectric charge.

BACKGROUND OF THE INVENTION

(2) The use of powder coating materials for the sealing and coloring of finishing surfaces is popular in many industrial and manufacturing processes. The use of powder coating materials can allow improved control over the application of paint to a surface. Furthermore, when combined with electrostatic process such as a triboelectric process, the powder coating is electrically attracted to the surface being coated allowing for an improved and consistent application. Powder coating has a longer lifespan than regular paint.

(3) Although numerous approaches have been applied to powder coating, it becomes challenging when the paint is applied to small objects. Current powder coat head designs often require nozzles that are 8-12 inches long in order for rifling in the nozzle to create an adequate vortex of powder coating before it is expelled from the nozzle. This makes such designs impractical for use on small parts or other objects. Considerable amounts of powder coating can be lost to the surrounding air. It would be highly desirable to have a nozzle design that could impart a vortex or similar flow of powder coating while allowing the nozzle itself to be sized for paint application on smaller parts.

SUMMARY

(4) Disclosed herein are implementations of a paint nozzle for use in the application of powder

coating on small parts.
(5) In one embodiment, the paint nozzle includes a nozzle body defining a nozzle body axis. The nozzle body includes one or more nozzle body sidewalls, a nozzle engagement upper surface, a nozzle dispensing lower surface and an internal nozzle chamber formed in between the upper and lower surfaces. A center dispensing channel connects the internal nozzle chamber to the dispensing lower surface to allow powder coating paint to be ejected from the paint nozzle. The center dispensing channel is orientated parallel and coincident with the nozzle body axis. Two or more vortex dispensing channels are spaced axially outward of the center dispensing channel and also connect the internal nozzle chamber to the dispensing lower surface to allow powder coating paint to be ejected from the paint nozzle. The two or more vortex dispensing channels are orientated at a dispensing angle relative to the nozzle body axis to generates a vortex flow of the powder coating around the center dispensing channel to improve application on a paint surface.
(6) In one embodiment, the dispensing angle is 15 degrees relative to the nozzle body axis. In another, the dispensing angle may range from 7.5 degrees to 75 degrees relative to the nozzle body axis. In one embodiment each of the two or more vortex dispensing channels include a bevel formed on the nozzle dispensing lower surface to further improve the flow of the powder coating out of the paint nozzle. In one embodiment, the two or more vortex dispensing channels may comprise three vortex dispensing channels. In one embodiment, the two or more dispensing channels may be aligned to generate a counter clockwise vortex in the powder coating as it is ejected from the paint nozzle.
(7) In one embodiment, the paint nozzle may be formed from a triboelectric material to impart an electrostatic charge to the powder coating prior to it being ejected from the paint nozzle. In one embodiment this is achieved through the use of virgin PTFE material to form the paint nozzle. In another embodiment, recycled PTFE may be utilized.
(8) In one embodiment, it is contemplated that the paint nozzle comprises a cylindrical body with a diameter of less than 60 mm and preferably 28 mm. In one embodiment, the one or more nozzle sidewalls have a sidewall height of less than 60 mm and preferably 28 mm. In one embodiment, the nozzle engagements surface comprises a cylindrical engagement surface of less than 60 mm and preferably 32 mm.
(9) In one embodiment, the internal nozzle chamber comprises a concave semi-spherical internal nozzle chamber. In one embodiment, the concave semi-spherical internal nozzle chamber has a diameter of less than 40 mm and preferably 22 mm. In one embodiment, the two or more vortex dispensing channels have a diameter of approximately 6 mm.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) In the detailed description presented below, reference will be made to the following drawings.
(2) FIG. **1** illustrates a perspective view of an embodiment of the present disclosure, e.g., a paint nozzle.
(3) FIG. **2** is a top view of the paint nozzle illustrated in FIG. **1**.
(4) FIG. **3** is a bottom view of the paint nozzle illustrated in FIG. **1**.
(5) FIG. **4** is a cross-sectional view of the paint nozzle illustrated in FIG. **1**.
(6) FIG. **5** illustrates an embodiment of the present disclosure, e.g., a paint nozzle.
(7) FIG. **6** is a bottom view illustration of the paint nozzle illustrated in FIG. **5**.
(8) FIG. **7** illustrates an embodiment of the present disclosure, e.g., a paint nozzle.
DETAILED DESCRIPTION
(9) Referring now to FIGS. **1-4**, which are exemplary illustrations of a paint nozzle **10** in accordance with an aspect of the current disclosure. The paint nozzle **10** is designed for the

distribution of paint onto part surfaces. In one aspect, it is contemplated that the paint nozzle **10** is a powder coating paint nozzle for the application of powder coating paints.

(10) The paint nozzle **10** includes a nozzle body **12** defining a nozzle body axis **14** (see FIG. **4**). The nozzle body **12** may be formed in any suitable manufacturing process such as milling, casting, printing, molding, etc. Although a variety of different materials may be utilized to form the nozzle body **12**, one aspect of the disclosure contemplates the use of triboelectric materials. Triboelectric materials impart an electric charge to materials passing through or past them. This is a significant advantage when utilized with powder coating paints. The electric charge imparted into the powder coating paints makes them attracted to the parts they are being applied to. This increases both the efficiency and well as the accuracy of the paint application. In one aspect, it is contemplated that virgin PTFE material may be utilized to form the nozzle body **12** as it operates efficiently as a triboelectric material. In another aspect, recycled PTFE may be utilized to form the nozzle body **12**.

(11) The nozzle body **12** may include one or more nozzle sidewalls **16**. In the aspect shown in FIGS. **1-4**, the nozzle body **12** is cylindrical and therefore has only a single sidewall **16**. In other contemplated aspects, however, any suitable shape and corresponding number of sidewalls may be utilized. The nozzle body **12** may include a nozzle engagement upper surface **18**. It is contemplated that the nozzle engagement upper surface **18** is configured to engage a paint gun or other application device. In one aspect, it is contemplated that the nozzle engagement upper surface **18** may be threaded or include other attachment features to securely attach to a paint gun or other application device. The nozzle body **12** also includes a nozzle dispensing lower surface **20**. An internal nozzle chamber **22** is formed within the nozzle body **12** and is positioned between the nozzle engagement upper surface **18** and the nozzle dispensing lower surface **20**. The internal nozzle chamber **22** allows the powder coating paint to mix within the nozzle body **12** and assists in imparting the electrical charge into the powder coating paint prior to application.

(12) In one aspect of the disclosure, it is contemplated that the internal nozzle chamber **22** may be formed as a concave semi-spherical chamber. This shape of the internal nozzle chamber **22** improves mixing and charging of the powder coating. In one aspect, the nozzle body **12** may include two or more vortex dispensing channels **24** spaced radially outward of the nozzle body axis **14**. In one aspect the two or more vortex dispensing channels **24** are contemplated to comprise three vortex dispensing channels **24**. The two or more vortex dispensing channels **24** each connect the internal nozzle chamber **22** to the nozzle dispensing lower surface **20** to allow power coating paint to pass from the internal nozzle chamber **22** outward through the nozzle dispensing lower surface **20**. Each of the two or more vortex dispensing channels **24** are orientated at a dispensing angle **26** relative to the nozzle body axis **14**. In one aspect, each of the two or more vortex dispensing channels **24** have identical dispensing angles **26** and relative orientation around the nozzle body axis **14**. This allows powder coating paint to generate a rotating vortex as it exits the nozzle dispensing lower surface **20**. In one aspect of the disclosure, it is contemplated that the two or more vortex dispensing channels **24** are orientated to generate a counter-clockwise vortex in the powder coating paint as it exits the nozzle dispensing lower surface **20**. In another aspect, a clockwise orientation is contemplated.

(13) It is contemplated that a wide variety of dispensing angles **26** may be utilized. In one aspect, a dispensing angle **26** of approximately 15 degrees was found to be beneficial to paint application. In other aspects, dispensing angles **26** between 7.5 degrees and 75 degrees were contemplated. It is further contemplated that the dispensing angle **26** may be tailored to accommodate the desired width of a paint application surface, the desired output of paint, and/or the distance between the surface to be painted and the nozzle body **12**. In another aspect, it is contemplated that each of the two or more vortex dispensing channels **24** may include a bevel **28** formed between the internal nozzle chamber and the nozzle dispensing lower surface **20**. It is contemplated that the bevel **28** is defined by a bevel angle **29** relative to the direction of the dispensing channels **24**. In one aspect, the bevel angle is less than 4 degrees. In another aspect, the bevel angle is 3 degrees. The addition

of bevels **28** allows for an improved vortex generation in the dispersed powder coating paint.

(14) In another aspect of the disclosure, the nozzle body **12** may include a center dispensing channel **30** connecting the internal nozzle chamber **22** to the nozzle dispensing lower surface **20** to allow power coating paint to pass from the internal nozzle chamber **22** outward through the nozzle dispensing lower surface **20**. In one aspect, the center dispensing channel **30** is orientated parallel and coincident with the nozzle body axis **14**. It is contemplated that the center dispensing channel **30** may include a bevel **28** formed at the nozzle dispensing lower surface **20**. The center bevel **28** may have a bevel angle of less than 4 degrees. In another aspect, the center bevel **28** may have a bevel angle of 3 degrees. When used in conjunction with the two or more vortex dispensing channels **24**, the center dispensing channel **30** provides a more concentrated paint application. In other aspects of the disclosure (FIGS. **6** and **7**), the two or more vortex dispensing channels **24** may be utilized without the center dispensing channel **30**.

(15) The present disclosure describes a paint nozzle **10** that can generate a vortex paint spray in a small, concentrated package. This allows it to be utilized on small parts and in smaller applications that conventional rifling induced spray nozzles may be unsuitable for. In one aspect of the disclosure, it is contemplated that the nozzle body **12** comprises a cylindrical body with a diameter less than 60 mm. In another aspect of the disclosure, it is contemplated that the nozzle body **12** comprises a cylindrical body with a body diameter **32** of approximately 28 mm. Similarly, in one aspect of the disclosure it is contemplated that the nozzle body sidewalls **16** may compromise a sidewall height **34** of less than 60 mm. In another aspect of the disclosure, it is contemplated that the sidewall height is approximately 28 mm. In another aspect of the disclosure, the nozzle engagement upper surface **18** is contemplated to be a cylindrical nozzle engagement upper surface **18** with an upper surface diameter **36** less than 60 mm. In another aspect of the disclosure, it is contemplated that the upper surface diameter **36** is approximately 32 mm. Similarly, in one aspect of the disclosure the internal nozzle chamber **22** may be formed as a concave semi-spherical chamber having a spherical diameter **38** of less than 40 mm. In another aspect of the disclosure, it is contemplated that the spherical diameter **38** is approximately 22 mm. Additionally, each of the two or more vortex dispensing channels **24** and the center dispensing channel **30** may comprise a channel diameter **40**. It is contemplated that the channel diameter is less than 15 mm and in one aspect is 6 mm.

(16) This disclosure is an improvement to the nozzles used in the "guns" that are used for this powder coating process. The disclosure describes a nozzle of compact design that creates a vortex in the powder to allow greater electrostatic charge over existing systems. Current powder coat head designs might require large and long nozzles which are not ideal in certain manufacturing environments. This disclosure describes a paint nozzle **10** that provides more efficient powder coating by way of vortex creation that results in more powder being deposited on the target and less powder wasted in the air. Testing indicated a possible 20-30% reduction in the amount of powder needed to achieve adequate coating of a product. It is contemplated that the described paint nozzle **10** is suitable for stationary or handheld paint gun systems.

(17) Any portion of the systems, apparatuses, methods, and processes herein may occur in any order or sequence. Certain components or steps may occur simultaneously, others may be added, and/or others may be omitted. This disclosure is provided for the purpose of illustrating certain embodiments and should in no way be construed so as to limit the claims.

(18) Accordingly, it is to be understood that the above description is intended to be illustrative and not restrictive. Many embodiments and applications other than the examples provided would be apparent upon reading the above description. The scope should be determined, not with reference to the above description, but should instead be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. It is anticipated and intended that future developments will occur in the technologies discussed herein, and that the disclosed systems and methods will be incorporated into such future embodiments. The

embodiments of this disclosure are capable of modification and variation.
(19) All terms used in the claims are intended to be given their broadest reasonable constructions and their ordinary meanings as understood by those knowledgeable in the technologies described herein unless an explicit indication to the contrary is made herein. Use of the singular articles such as “a,” “the,” “said,” etc. should be read to recite one or more of the indicated elements unless a claim recites an explicit limitation to the contrary.
(20) The Abstract of the Disclosure is provided to allow the reader to ascertain the nature of the technical disclosure, but it should not be used to interpret or limit the scope or meaning of the claims. Various features of this disclosure may be grouped together in various embodiments for the purpose of streamlining the disclosure, but the claimed embodiments shall not be interpreted as requiring more features than are expressly recited in each claim. The inventive subject matter of the claims lies in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.
(21) It will be appreciated by those skilled in the art that modifications can be made to the embodiments disclosed and remain within the inventive concept. Therefore, this invention is not limited to the specific embodiments disclosed, but is intended to cover changes within the scope and spirit of the claims.

Claims

1. A paint nozzle comprising: a nozzle body defining a nozzle body axis, the nozzle body including one or more nozzle sidewalls, a nozzle engagement upper surface, a nozzle dispensing lower surface, and an internal nozzle chamber formed in between the nozzle dispensing lower surface and the nozzle engagement upper surface; a center dispensing channel connecting the internal nozzle chamber to the nozzle dispensing lower surface, the center dispensing channel orientated parallel and coincident with the nozzle body axis; two or more vortex dispensing channels spaced radially outward from the center dispensing channel, each of the two or more vortex dispensing channels connecting the internal nozzle chamber to the nozzle dispensing lower surface, the two or more vortex dispensing channels orientated at a dispensing angle relative to the nozzle body axis.
2. The paint nozzle of claim 1, wherein the dispensing angle is 15 degrees relative to the nozzle body axis.
3. The paint nozzle of claim 1, wherein the dispensing angle is between 7.5 degrees and 75 degrees.
4. The paint nozzle of claim 1, wherein each of the two or more vortex dispensing channels includes an exit bevel formed between the internal nozzle chamber and the nozzle dispensing lower surface.
5. The paint nozzle of claim 1, wherein the nozzle body comprises a triboelectric material.
6. The paint nozzle of claim 5, wherein the nozzle body comprises one of virgin PTFE and recycled PTFE.
7. The paint nozzle of claim 1, wherein the two or more vortex dispensing channels comprise three vortex dispensing channels.
8. The paint nozzle of claim 1, wherein the nozzle body comprises a cylindrical body with a diameter of approximately 28 mm.
9. The paint nozzle of claim 1, wherein the one or more nozzle sidewalls have a sidewall height of approximately 28 mm.
10. The paint nozzle of claim 1, wherein the nozzle engagement surface comprises a cylindrical engagement surface with a diameter of approximately 32 mm diameter.
11. The paint nozzle of claim 1, wherein the two or more vortex dispensing channels are orientated to generate a vortex in a counterclockwise direction.
12. The paint nozzle of claim 1, wherein the internal nozzle chamber comprises a concave semi-

spherical internal nozzle chamber.
13. The paint nozzle of claim 12, wherein the concave semi-spherical internal nozzle chamber comprises a spherical diameter of approximately 22 mm.
14. A powder coating paint nozzle comprising: a nozzle body defining a nozzle body axis, the nozzle body including one or more nozzle sidewalls, a nozzle engagement upper surface, a nozzle dispensing lower surface, and an internal nozzle chamber formed in between the nozzle dispensing lower surface and the nozzle engagement upper surface; and two or more vortex dispensing channels spaced radially outward from the nozzle body axis, each of the two or more vortex dispensing channels connecting the internal nozzle chamber to the nozzle dispensing lower surface, the two or more vortex dispensing channels orientated at a dispensing angle relative to the nozzle body axis, each of the two or more vortex dispensing channels including an exit bevel formed between the internal nozzle chamber and the nozzle dispensing lower surface.
15. The powder coating paint nozzle of claim 14, wherein the exit bevel has a bevel angle of less than four degrees.
16. The powder coating paint nozzle of claim 14, wherein the exit bevel has a bevel angle of approximately three degrees.
17. The powder coating paint nozzle of claim 14, wherein the nozzle body comprises a cylindrical body with a diameter less than 60 mm.
18. The powder coating paint nozzle of claim 14, wherein the internal nozzle chamber comprises a concave semi-spherical internal nozzle chamber having a spherical diameter of less than 30 mm.